WORKSHEET 7.5

Separation of Variables

Unit 7: Differential Equations

Key ideas

- If $\dfrac{dy}{dx} = g(x)\,h(y)$, move everything in y to one side and everything in x to the other: $\dfrac{dy}{h(y)} = g(x)\,dx$.
- Integrate both sides. Add **one** constant of integration, at this step.
- Then solve for y. An exponential turns the constant into a multiplier: $y = Ae^{\cdots}$.
- Dividing by $h(y)$ can lose a constant solution where $h(y) = 0$. Check for it separately.

Worked examples

Example 1 · An exponential solution

Solve $\dfrac{dy}{dx} = 3x^2 y$.

Solution. $\dfrac{dy}{y} = 3x^2\,dx$, so $\ln|y| = x^3 + C$.

$y = Ae^{x^3}$.

Example 2 · A square root

Solve $\dfrac{dy}{dx} = \dfrac{\cos x}{y}$.

Solution. $y\,dy = \cos x\,dx$, so $\dfrac{y^2}{2} = \sin x + C_0$.

$y^2 = 2\sin x + C$.

Example 3 · A reciprocal

Solve $\dfrac{dy}{dx} = \dfrac{y^2}{x}$ for $x > 0$, $y \neq 0$.

Solution. $y^{-2}\,dy = \dfrac{dx}{x}$, so $-\dfrac{1}{y} = \ln x + C$.

$y = -\dfrac{1}{\ln x + C}$.

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Practice

CORE

1. Separate the variables in $\frac{dy}{dx} = xy^3$. Do not integrate.

2. Solve $\frac{dy}{dx} = 4y$.

3. Solve $\frac{dy}{dx} = 6x^2$.

4. Solve $\frac{dy}{dx} = -y$.

EXAM LEVEL

5. Solve $\frac{dy}{dx} = \frac{x+1}{y}$, leaving your answer in the form $y^2 = \ldots$

6. Solve $\frac{dy}{dx} = y \sin x$.

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7. Solve $\dfrac{dy}{dx} = e^{2x-y}$.

8. Solve $\dfrac{dy}{dx} = \sqrt{1-y^2}$ for $-1 < y < 1$.

Question 9 · Free-response practice

Consider the differential equation $\dfrac{dy}{dx} = x\sqrt{y}$ for $y \geq 0$.

(a) Separate the variables.

(b) Find the general solution for $y > 0$, in the form $y = \ldots$

(c) Show that $y = 0$ is also a solution, and explain why the method of part (b) did not produce it.

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Worked solutions

1. $\dfrac{dy}{y^3} = x\,dx$.

2. $y = Ae^{4x}$.

3. $y = 2x^3 + C$.

4. $y = Ae^{-x}$.

5. $y\,dy = (x+1)\,dx$: $y^2 = x^2 + 2x + C$.

6. $\ln|y| = -\cos x + C$: $y = Ae^{-\cos x}$.

7. $e^y\,dy = e^{2x}\,dx$: $e^y = \dfrac{1}{2}e^{2x} + C$, so $y = \ln\left(\dfrac{1}{2}e^{2x} + C\right)$.

8. $\dfrac{dy}{\sqrt{1-y^2}} = dx$: $\arcsin y = x + C$, so $y = \sin(x + C)$.

9. (a) $\dfrac{dy}{\sqrt{y}} = x\,dx$.

   (b) $2\sqrt{y} = \dfrac{x^2}{2} + C_0$, so $y = \left(\dfrac{x^2}{4} + C\right)^2$.

   (c) If $y = 0$ for all x, both sides of the equation are 0. Separating required division by $\sqrt{y}$, which is not allowed when $y = 0$.

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